DHAYALAN K

Cybersecurity & Systems Security Engineer | Protocol Analysis, Host Hardening & CTF Defense

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IMPACT AT A GLANCE

3 National CTF Challenges Tamil Nadu Police Yukthi CTF 2.0, Amrita Lemon CTF, REC HackTiVate 24h CTF | **Protocol Reverse Engineering** Z-Protocol raw socket transport with 16-byte binary framing & CRC32 integrity | **Host Defense & Hardening** automated Ansible UFW default-deny firewall policies & SSH root disablement | **Encrypted Traffic Research** statistical cross-protocol packet metadata leakage and timing analysis

Cybersecurity & Systems Security specialist with competitive hands-on experience in 24-hour Capture The Flag (CTF) challenges (Tamil Nadu Police Yukthi CTF 2.0, Amrita Lemon CTF, REC HackTiVate), low-level network protocol analysis, and Linux host defense. Experienced in raw socket reverse engineering, cryptographic handshake validation, encrypted traffic pattern analysis, and automated Linux server hardening (UFW, SSH, iptables) via Ansible. Driven by a deep understanding of transport layers, binary frame parsing, and defensive systems engineering.

SECURITY RESEARCH & SYSTEMS DEFENSE

Network Protocol Security & Traffic Analysis

- Authored **Cross-Protocol Traffic Fingerprinting** research: investigated side-channel metadata leakage, packet burst intervals, and traffic signatures across protocol boundaries.
- Built Python feature extraction pipelines analyzing packet timing distributions and frame size entropy to classify encrypted network streams without decrypting payload contents.
- Engineered **Z-Protocol Wire Transport**: from-scratch Layer-4 binary networking protocol on raw Python sockets with a custom 16-byte header, dynamic control flags (SYN, ACK, FIN, RST, PSH), and strict state machine transitions.
- Implemented real-time CRC32 checksum calculation and bitmask verification to detect packet tampering and sequence replay attacks.

Linux Host Hardening & Secure Remote Access

- Automated Linux server security baselines using Ansible: enforced UFW firewall default-deny policies, restricted SSH attack surfaces (key-only auth, root login prohibition), and automated audit logs.
- Architected **ONey Secure Remote Access**: encrypted peer-to-peer UDP hole punching for host administration without centralized relay vulnerabilities.

TECHNICAL SKILLS

Security & Network Analysis: Wireshark, Nmap, TCPDump, Raw Sockets (AF_INET/SOCK_RAW), Traffic Fingerprinting, Packet Metadata Analysis, Protocol Reverse Engineering

Host Hardening & Defense: Linux Security (Ubuntu/Debian), UFW Firewall Automation, SSH Hardening (Disable root login, key-only auth), Systemd Sandboxing, File Permissions

Programming & Scripting: Python (Security Tools, Socket Programming), Java (Secure OOP, Encapsulation — 92% Merit), Bash Scripting

Protocols & Cryptography: TCP/IP State Machines, UDP Hole Punching, CRC32 Checksums, Symmetric/Asymmetric Keys, Binary Framing, TLS/SSL Certificates

CTF Domains & Exploitation: Network Forensics, Web Security (OWASP Top 10), Cryptography Challenges, Linux Privilege Escalation, Binary Inspection

PROFESSIONAL EXPERIENCE

Python & Security Tools Builder, CTF Competitions & Systems Lab **2024 to Present, Coimbatore / Chennai**

- Competed in 24-hour national CTFs: Tamil Nadu Police Yukthi CTF 2.0, Amrita L3m0n CTF, and Rajalakshmi HackTiVate.
- Constructed custom network protocol inspectors and packet metadata extraction tools in Python.
- Investigated defensive countermeasures against encrypted traffic analysis in peer-to-peer architectures.

Systems Defense & Automation Developer, Independent Infrastructure Systems **2025 to Present, Coimbatore**

- Engineered automated Ansible security playbooks enforcing UFW default-deny policies across distributed server nodes.
- Developed ONey encrypted P2P daemon with session-key verification to prevent unauthorized remote execution.

KEY PROJECTS & CODEBASES

Cross-Protocol Traffic Fingerprinting Research github.com/DHAYALANKANIAPPAN/PROJECTS

- Analyzed packet metadata, burst timings, and entropy distributions to classify protocols across encrypted streams.
- Evaluated classification accuracy in encrypted environments. Stack: Python, Traffic Analysis, Packet Metadata.

Z-Protocol Wire Transport & Handshake State Machine github.com/DHAYALANKANIAPPAN/Z-Protocol

- Designed Layer-4 binary networking protocol on raw Python sockets with custom 16-byte header and CRC32 verification.
- Implemented 3-way handshake state machine (SYN/SYN-ACK/ACK) and packet reordering safety. Stack: Python, Raw Sockets, CRC32.

ONey Secure Remote Access & Encrypted P2P Tunneling github.com/DHAYALANKANIAPPAN/PROJECTS

- Low-latency remote system management client with encrypted P2P UDP tunneling for host administration without relays.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering (Cyber Security), Pursuing

Sri Shakthi Institute of Engineering and Technology, Coimbatore (2024 – 2028) | Grade Point Average: 8.21 out of 10.0. Available for full-time employment & internships.

VERIFIED CERTIFICATIONS, AWARDS & ACTIVITIES

- Yukthi CTF 2.0 (National-Level CTF) — Tamil Nadu Police Cyber Crime Wing & Selfmade Ninja Academy (Digital Resilience).
- 0xTi - HackTiVate 24-Hour CTF (Titanium 2026) — Rajalakshmi Engineering College (Binary analysis & web exploitation).
- L3m0n CTF 2025 (24-Hour CTF) — Amrita Vishwa Vidyapeetham (Cryptography & network forensics).
- Amazon ECS Knowledge Badge Assessment — AWS Credly Verified (Container isolation & IAM task roles).
- Certificate of Merit: Java + MongoDB (92% Score) — Hazhtech Career Development Centre / IACT.